

KARGO — B2B warehouse-to- warehouse freight, mapped end to end

Actors, chain of custody, order registration flow and data model of a chain-of-custody freight platform, prepared for an external logistics review before development starts.

What we are asking of the consultant

The digital flow is defined. What is **not** validated is the operational reality behind it. Section 8 lists seven questions where a logistics practitioner's judgement decides how the system must behave (failed scans, partial deliveries, label production, SLAs, returns, multi-client operators). Sections 2–7 are the context you need to answer them.

Document date

21 September 2026

Status

Design phase — no code written yet

Market

Chile • intercity bus line-haul

01 Executive summary

KARGO coordinates freight between a client's own warehouses. A **Merchant** creates an order; a **Logistics Coordinator** assigns it to a **Transport Operator**, which collects, consolidates, line-hauls by intercity bus and delivers to the destination warehouse.

The business model

- B2B only, warehouse to warehouse, one client account per relationship.
- No payment at order creation. Billing is consolidated monthly per client.
- **ON HOLD** The billing calculation basis has not been communicated yet and is parked to the end of the project.

The core design idea

- Every physical hand-over of a package is a **control point**.
- Each control point records **who, when** (date + time) and a **scan of every package**.
- Pickup and final delivery additionally capture an **electronic signature** from a pre-registered authorized person.

Scope of the first release

- Merchant creates orders and tracks them (read-only) — from a phone.
- Coordinator assigns; operator crews execute with a mobile scanning app.
- Backend is simulated with static JSON files for the prototype.

How the two reference apps informed this

- **kargo-flow-2**: strong visuals, working mobile order form, but no package data.
- **kargo 2.0**: strong process vision (multi-role, per-package manifest, event feed) but a demo simulation, not mobile-usable.
- Decision: rebuild, combining the best of both.

Design rules that govern every screen

1. The Merchant **only creates** orders. It never assigns carriers, scans, or manages operations.
2. Mobile first — every role must be able to work from a phone.
3. Every hand-over: scan of each package + timestamp + responsible user, with no exceptions.
4. Identity verification is **visual only**: the Merchant registers at least two authorized persons per handover point (name, RUT, phone); the operator compares the RUT with the physical ID.
5. Registration must take the fewest possible taps — it is the highest-frequency action in the system.

02 Actors and organizational levels

Three organizational levels and seven roles. The Merchant deliberately sees none of the operational complexity below the first step.

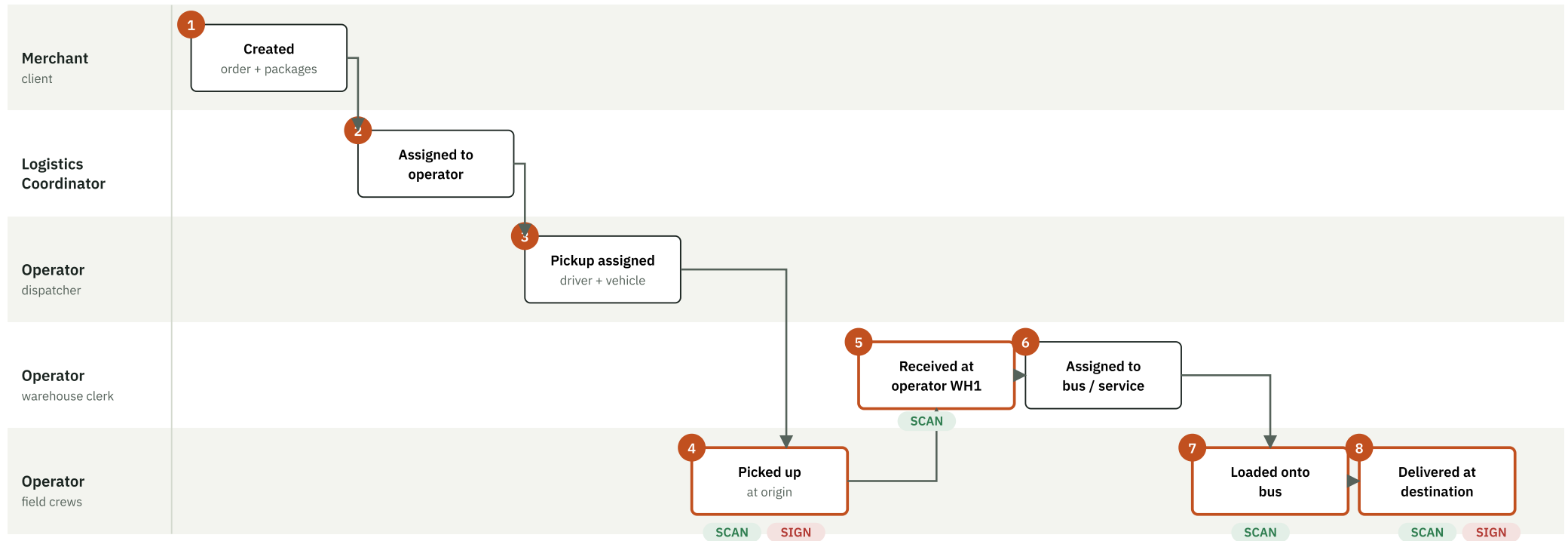
ROLE	LEVEL	RESPONSIBILITY	SCANS	SIGNS
Merchant	Client	Creates the order: warehouses, authorized persons, packages. Read-only tracking afterwards.	—	—
Logistics Coordinator	Kargo	Receives created orders; assigns each to a Transport Operator.	—	—
Dispatcher	Operator	Receives the assignment; designates the pickup driver and vehicle.	—	—
Pickup Driver / Assistant	Operator	Collects at the origin warehouse.	Every package	Yes (origin)
Operator Warehouse Clerk (WH1)	Operator	Receives at the operator's warehouse; assigns the load to a bus / service.	Every package	—
Bus Driver / Assistant	Operator	Confirms loading of each package onto the bus.	Every package	—
Delivery Driver / Assistant	Operator	Delivers to the destination warehouse.	Every package	Yes (destination)

Merchant-facing status is intentionally simplified: Created → Assigned → Pickup in progress → In transit → Delivered. The eight internal steps (next section) exist in the back office and in the operator app only.

What each level is not allowed to do

- **Merchant** cannot assign an operator, scan packages, or edit operational data.
- **Operator crews** cannot create or modify the order's commercial data (warehouses, package declaration).
- No role can alter a timestamp or delete a scan; corrections are new, attributed events.

03 Chain of custody — eight steps, four control points



Orange-outlined boxes are physical hand-overs (steps 4, 5, 7, 8). Every step also stores the acting user and a date-time stamp.

SIGN = electronic signature of a pre-registered authorized person, RUT compared visually to physical ID (steps 4 and 8 only).

The Merchant sees only a five-state summary of this chain.

Swimlane view of an order from creation to closure at the destination warehouse.

04 Control points and captured evidence

What the system records at each step. This is the audit trail that will settle disputes ("24 of 25 packages arrived") and, later, support the monthly invoice.

#	STEP	ACTOR	EVIDENCE CAPTURED
1	Created	Merchant	USER + TIME Origin/destination, ≥2 authorized persons per point, package declaration. One printable code per package is generated.
2	Assigned to operator	Coordinator	USER + TIME Operator id.
3	Pickup assigned	Dispatcher	USER + TIME Driver id and vehicle plate.
4	Picked up	Pickup crew	SCAN / PACKAGE SIGNATURE USER + TIME Person handing over is one of the pre-registered authorized persons; RUT visually checked.
5	Received at operator WH1	Warehouse clerk	SCAN / PACKAGE USER + TIME
6	Assigned to bus / service	Warehouse clerk	USER + TIME Bus plate, service name.
7	Loaded onto bus	Bus crew	SCAN / PACKAGE USER + TIME Bus driver id.
8	Delivered	Delivery crew	SCAN / PACKAGE SIGNATURE USER + TIME Receiver is one of the pre-registered authorized persons; RUT visually checked. Order closed at destination.

Metrics that this design makes computable for free

Because every hand-over is timestamped and attributed, the following can be derived without extra data capture. These are **candidate** indicators offered for the consultant to confirm or replace:

Time-based

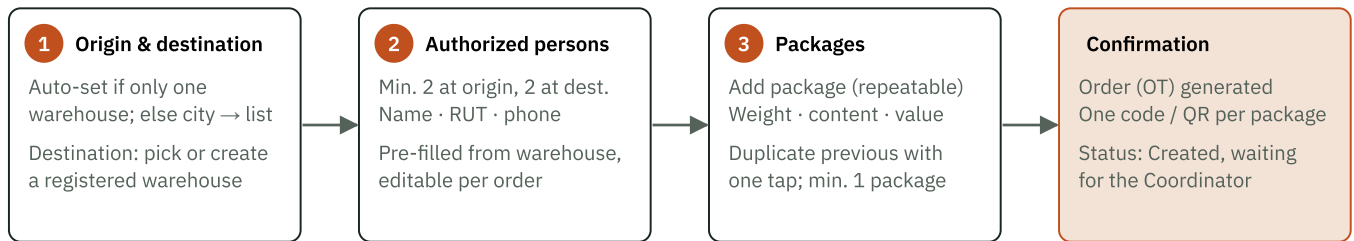
- Creation → assignment (coordinator responsiveness)
- Assignment → pickup (operator lead time)
- Dwell time at operator WH1 (step 5 → step 7)
- End-to-end lead time vs. SLA

Integrity-based

- Scan completeness per control point (packages scanned ÷ packages declared)
- Partial delivery rate and where the gap first appears
- Handovers by person / crew (accountability)
- Forced-manual registrations vs. real scans

05 Merchant order registration flow

A three-step, phone-first wizard. The Merchant's work ends at confirmation; everything in section 3 happens afterwards.



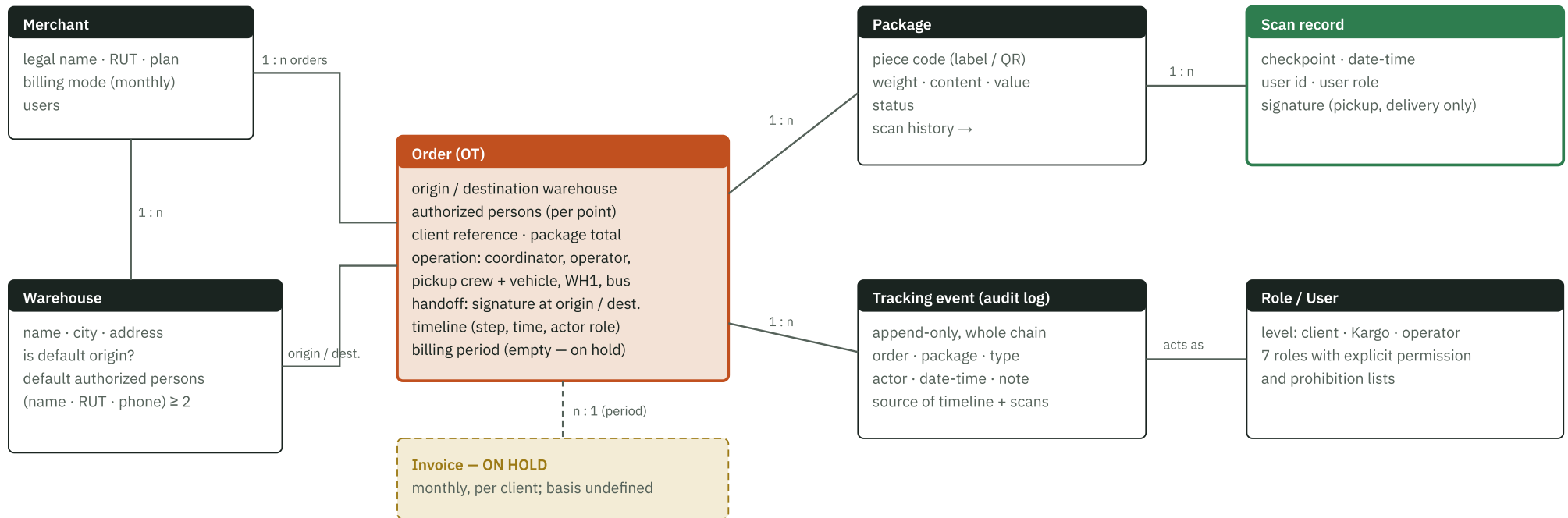
Rules that shape the flow

- **No scan or photo is required from the Merchant.** The mandatory scans (and signature) are performed later by operator crews at each hand-over.
- **The only identity check is visual.** The Merchant supplies the authorized persons; the crew member checks the physical ID against the RUT on screen.
- **Warehouses are registered once** per Merchant account and reused. Most clients will have a single warehouse, in which case origin selection disappears; the model supports many, filtered by city.
- **High-volume path:** Create → Bulk upload (CSV template, origin fixed per account, row-level error preview before confirmation).
- **No payment step.** Billing is monthly and consolidated.

Merchant portal sections

SECTION	PURPOSE
Home	Daily indicators: active, delivered, incidents; shortcut to register a shipment.
Orders	List, search, filter by the five-state summary status.
Tracking	Order detail: simplified timeline, per-package status, signatures once captured.
Incidents / Returns	Reports and return requests linked to a specific package.
Documents	Manifests and package labels (PDF).
Account	Registered warehouses (with city), default authorized persons.
Billing ON HOLD	Monthly consolidated view — to be defined.

06 Data model



Entity view. In the prototype each entity is a static JSON file (merchants, orders, packages, tracking_events, roles, invoices) acting as the contract for the development team.

Design choices worth reviewing

- The order timeline and each package's scan history are **derived views** of an append-only event log.
- State is never overwritten; corrections are new attributed events.

Why per-package identity

- An order can contain 1 to 40+ packages.
- Scanning each one makes "which package is missing, and where did it disappear?" answerable from data.

07 System knowledge map

Automatically derived from the specification documents and data files: 61 concepts, 102 relationships, grouped into 8 clusters by connectivity (number = concepts in the cluster; green = data model). Used to check that nothing in the design is disconnected. The full interactive graph accompanies this document (graph.html).

Scan and signature custody rules 13

Chain of custody · pickup and delivery steps · scan + timestamp + user rule · visual ID check

Coordination and operator roles 11

Transport Operator · Coordinator · Dispatcher · crews · steps 1–3, 6

Product scope and principles 11

KARGO platform · Merchant · mobile-first · merchant only creates · billing (on hold)

Merchant order wizard 10

Wizard steps · confirmation · package code / QR · bulk upload · portal sections

Order and role data 5

Order entity · Role entity · orders.json · roles.json

Package and audit-trail data 5

Package · Scan record · Tracking event · packages / tracking_events.json

Merchant and warehouse data 4

Merchant · Warehouse · origin dropdown by city · merchants.json

Authorized persons verification 2

Authorized persons (name, RUT, phone) · open question: closed list vs free entry

Strongest links between clusters

- Custody rules ↔ Operator roles — **8 links** (the heart of the system)
- Operator roles ↔ Product scope — 3
- Order data ↔ Package / audit data — 2
- Order wizard ↔ Operator roles — 2 (creation feeds the chain)

What the map revealed

- The design is coherent: the scan rule ties together roles, steps and data.
- 9 concepts are weakly connected — nearly all are the **open operational questions** of section 8. These are the least-specified areas of the system.

08 Questions for the logistics consultant

The digital flow is specified. These seven points depend on operational experience, and each one changes what the software must do.

- 1 Failed or impossible scans.** What is the accepted procedure when there is no signal, the label is damaged or the device fails?

Proposal on the table: a manual forced registration with a mandatory reason code, flagged for review. Is that acceptable, and who reviews it?

- 2 Partial deliveries.** If 24 of 25 packages are scanned at destination, when is an order "closed", who is notified, and how does it affect the SLA clock?

Determines the incident workflow and the closure rule for step 8.

- 3 Label production.** Portable thermal printer at the client's warehouse, or labels printed in an office beforehand?

Affects hardware budget and how quickly a package can be tied to its order.

- 4 Authorized persons.** Closed list per warehouse or free entry per order? How many are really needed for a warehouse that is staffed by one person?

Two persons minimum is the current rule; it is the only identity control in the system.

- 5 Operator model.** Does one Transport Operator serve several clients, or are operators dedicated? Can the Coordinator re-assign an order after pickup?

Changes operator modeling and whether operators need their own coordinator view.

- 6 Service levels.** Which SLA targets apply per route and per service type (standard / express), and where does the clock start and stop?

The timestamps to compute it already exist; the targets do not.

- 7 Returns and damaged goods.** What evidence (photos at each scan?), what liability boundary between operator and client, and what return flow?

Photo capture is currently only implied at scan time; it may need to be explicit at every control point.

Deliberately out of scope for this review: billing. The monthly consolidated invoice exists in the model as an empty placeholder, and its calculation basis (per package, per order, per weight or fixed contract rate) is pending information from the business. It will be designed last and does not block any other phase.

09 Proposed delivery sequence

PHASE	DELIVERABLE	WHY IN THIS ORDER
1	Merchant registration wizard + order list (mobile-first, JSON backend)	Highest-frequency action; validates the data captured at creation.
2	Simplified tracking view + per-package status	What a client asks for first once orders exist.
3	Logistics Coordinator portal: pending orders, assign to operator	Closes the first hand-over of the chain.
4	Operator mobile scanning app for the four crew roles (camera scan, signature at pickup and delivery)	Core of traceability; without it there is no proof of hand-over.
5	Bulk CSV upload, incidents, returns	High-volume clients and exception handling.
6	Documents, reports / SLA, billing ON HOLD	Used occasionally; billing awaits business definition.

KARGO system map · prepared for external logistics review · design phase, no production code. Companion files: specification documents (docs/), JSON data contract (data/), interactive knowledge graph (graphify-out/graph.html).